GitGuardian Canary Tokens released to identify corrupted DevOps environments

A new open source canary tokens project has been launched, according to code security platform vendor GitGuardian, to aid enterprises in identifying corrupted developer and DevOps environments. The company claims that security teams may develop and deploy canary tokens in the form of Amazon Web Services (AWS) secrets to set off alarms as soon as they are tampered with by attackers. This can be done by using GitGuardian Canary Tokens (ggcanary).

According to a news release from GitGuardian, enterprises are inadvertently increasing their attack surfaces as a result of their continuing adoption of the cloud and contemporary software development techniques. Continuous integration and continuous deployment (CI/CD) pipelines are becoming popular entry vectors for attackers as a result of poorly protected business networks and internet-facing assets, it was noted.

After getting initial access, attackers frequently look for genuine hard-coded credentials they can exploit to migrate laterally. According to GitGuardian, the ggcanary project has many features to assist organisations in identifying compromises more quickly. These include: reliance on Terraform to build and manage AWS canary tokens using the well-known infrastructure-as-code application tool from HashiCorp; highly sensitive intrusion detection that tracks every action an attacker takes on the canary tokens using AWS CloudTrail audit records; and scalability of up to 5,000 active AWS canary tokens installed on the internal perimeter of a company, in ticketing, CI/CD tools, source-code repositories, and messaging platforms like Jira, Slack, or Microsoft Teams.

Cirq is now the common language for Google engineers to create quantum programs

After more than four years of development, the Google open source quantum programming framework Cirq has gained stability and established itself as the common language used by Google engineers to create quantum programs on Google's own quantum processors. As a result, upcoming point releases, like 1.1, will be compatible with the version that serves as its primary foundation, like 1.0. Breaking changes could appear in new major releases, such as 2.0.

Cirq is intended for today's noisy intermediate-scale quantum computers, which have at most a few thousand quantum gates and a few hundred qubits. Google claims that in order to produce cutting-edge results with contemporary quantum computers, one must deal with the specifics of their hardware setup, which explains the abstractions the framework offers.

This implies, for instance, that Cirq allows for the specification of the mapping between the algorithm and the hardware, offering fine-grained control down to the gate level, as well as dealing with specific processor constraints, which if not addressed properly, could lead to incorrect computations. A circuit, also referred to as a Cirq program, is a grouping of moments. Every operation in a given moment is the outcome of applying a gate to a disjoint subset of the available qubits. A moment is a collection of simultaneous operations.

Cirq has been enhanced since its debut in 2018 by a number of libraries investigating various facets of quantum computing research, including TensorFlow Quantum, OpenFermion, Pytket, and others. Specifically, a machine learning model has been trained on 30 qubits using TensorFlow Quantum at a pace of 1.1 petaflops per second. To mimic fermionic systems, such as quantum chemistry, OpenFermion provides a library for assembling and examining quantum algorithms.

Nilekani Centre at IIT Madras to develop open source AI for Indian languages

The Indian Institute of Technology Madras has launched the 'Nilekani Centre' as part of its AI4Bharat project to enhance Indian language technology. The Centre has received a donation of Rs 360 million from Nilekani Philanthropies, and was inaugurated by Nandan Nilekani.

IIT Madras launched AI4Bharat as a project to provide open source language AI for Indian languages. Modern methods for speech recognition and machine translation have been created here by a team led by Dr Mitesh M. Khapra, Pratyush Kumar, and Anoop Kunchukuttan.